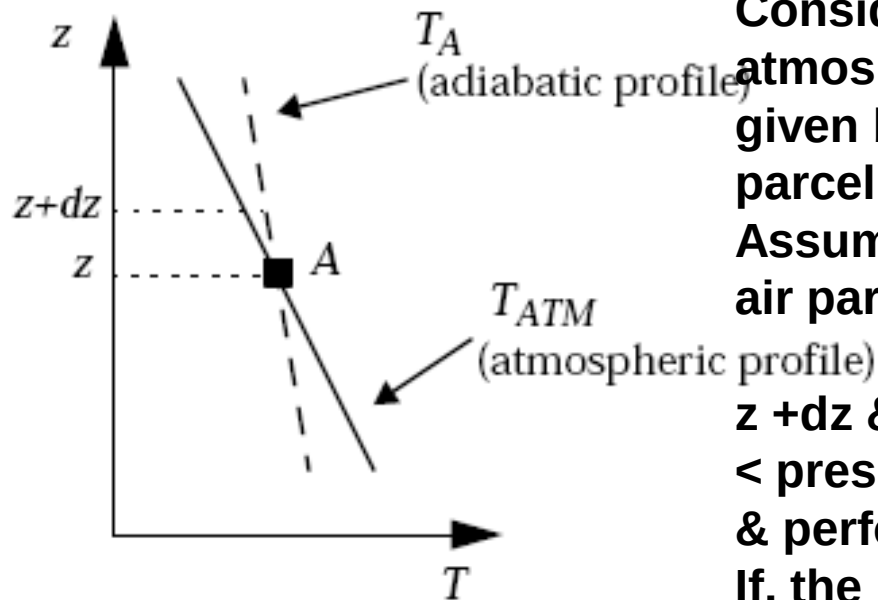


ATMOSPHERIC STABILITY

- We saw last time that Buoyancy in the atmosphere is driven by the vertical gradient of temperature



Consider a horizontally homogeneous atmosphere with vertical temp profile given by $T_{ATM}(z)$. Let A represent an air parcel at altitude z in this atmosphere. Assume that by some small ext. force, air parcel A is pushed upward from z to

$z+dz$ & then released. Pressure at $z+dz <$ pressure at z , so air parcel expands & performs work : $dW = -PdV$.

If, the rise of air parcel is adiabatic (i.e. no exchange of energy with surroundings (i.e. $dQ = 0$), then work is done at expense of internal energy E of air parcel &:
 $dE = dW + dQ$ (1st law) $= -PdV < 0$. As E (the internal energy depends only on T for ideal gas), it implies that air parcel cools. Cooling is shown as a dashed line in adj. Figure

ATMOSPHERIC STABILITY

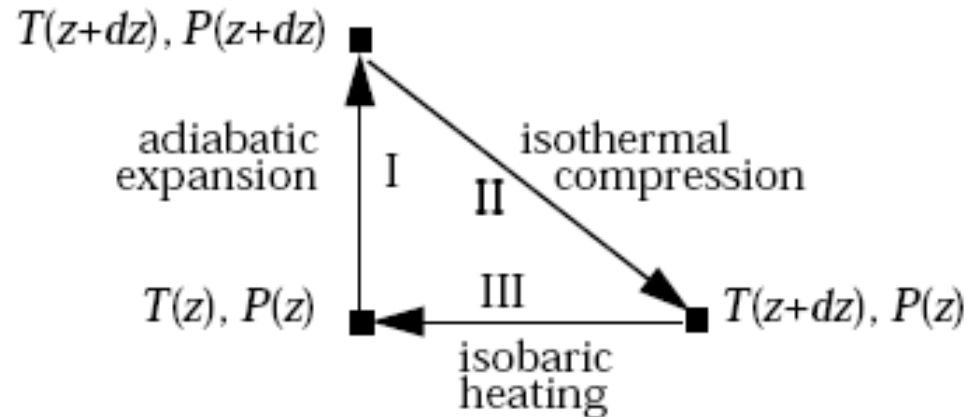
- One might expect that with cooling the air parcel gets heavier than its surroundings and therefore sinks back to its original position on account of buoyancy
- However, T_{ATM} also decreases with altitude
- The rise of air parcel depends on its “adiabatic cooling rate” relative to change in temp. with altitude.
- If $T_A(z+dz) > T_{\text{ATM}}(z+dz)$, the air parcel is warmer than surrounding air at same altitude. As a result density ρ , is less than ρ' (density of surrounding air) & air parcel will continue to get accelerated upward by buoyancy. **Implies Unstable Atmosphere w.r.t Vertical Motion...(Also termed as convective atmosphere)**
- On the contrary, if $T_A(z+dz) < T_{\text{ATM}}(z+dz)$, then rising air parcel is colder & heavier & sinks back; **Stable atmosphere w.r.t Vertical Motion, which is suppressed**

Criteria for Atmospheric Stability: Lapse rates

- Rate of decrease of temperature with altitude ($-dT/dz$) is called the **lapse rate**
- To determine whether an atmosphere is **stable or unstable** we need to compare its **atmospheric lapse rate** ($-dT_{ATM}/dz$) to the **adiabatic lapse rate** ($-dT_A/dz$)
- Note that “stability” is a **local** property of the atmosphere; an atmosphere may be stable at some altitudes and unstable at others
- Moreover “stability” refers to both upward and downward motions; if an atmosphere is unstable it holds for both upward and downward motions and hence causes rapid vertical mixing rather than unidirectional transport

Derivation of Adiabatic Lapse Rate

Consider the following thermodynamic cycle



Air parcel $[T(z), P(z)]$ rises adiabatically from z to $z + dz$ (process I), then compresses from $z + dz$ to z (process II), & finally heats isobarically at altitude z (process III)

The cycle returns the air parcel to its initial state, so must have zero net effect on any thermodynamic function.

Derivation of Adiabatic Lapse Rate

Consider the enthalpy (H) of the air parcel. It is defined by $H = E + PV$, where E is internal energy of air parcel

The change in enthalpy during any thermodynamic process is:

$$dH = dE + d(PV) = dW + dQ + d(PV)$$

Where, $dW = -PdV$ & dQ is heat added to the system.

Expanding $d(PV)$, we obtain

$$dH = -PdV + dQ + PdV + V dP = dQ + VdP$$

For the process I, discussed in the prev. Slide, $dQ = 0$ by definition (adiabatic) so that,

$$dH_I = VdP \quad \text{(Eqn 1)}$$

Derivation of Adiabatic Lapse Rate

- For the isothermal process II, $dE = 0$ (internal energy of an ideal gas is a function of temperature only) & $d(PV) = 0$ (ideal gas law), so that

$$dH_{II} = 0 \quad (\text{Eqn 2})$$

- For the isobaric process III,

$$dH_{III} = dQ = m C_p (T(z) - T(z+dz)) = -m C_p dT \quad (\text{Eqn 3})$$

where, m is mass of air parcel & $C_p = 1000 \text{ J/kg/K}$, the specific heat of air at constant pressure.

By definition of the thermodynamic cycle,

$dH_I + dH_{II} + dH_{III} = 0$, so that:

$$VdP = m C_p dT ; \text{ Recall } \frac{dP}{dz} = -\rho_a g \quad ; \quad \& \quad m = \rho_a V, \text{ so that,}$$

$$-dT/dz = g / C_p = 9.8 \text{ K/km} = \Gamma \text{ (Adiabatic lapse rate)}$$

Summary of Stability & Adiabatic Lapse Rate Relation

- Remarkably, Γ is a constant independent of atmospheric conditions. We can diagnose whether an atmosphere is stable or unstable with respect to vertical motions simply by comparing its lapse rate to $\Gamma = 9.8 \text{ K km}^{-1}$:

$$-\frac{dT_{ATM}}{dz} > \Gamma \quad \text{unstable}$$

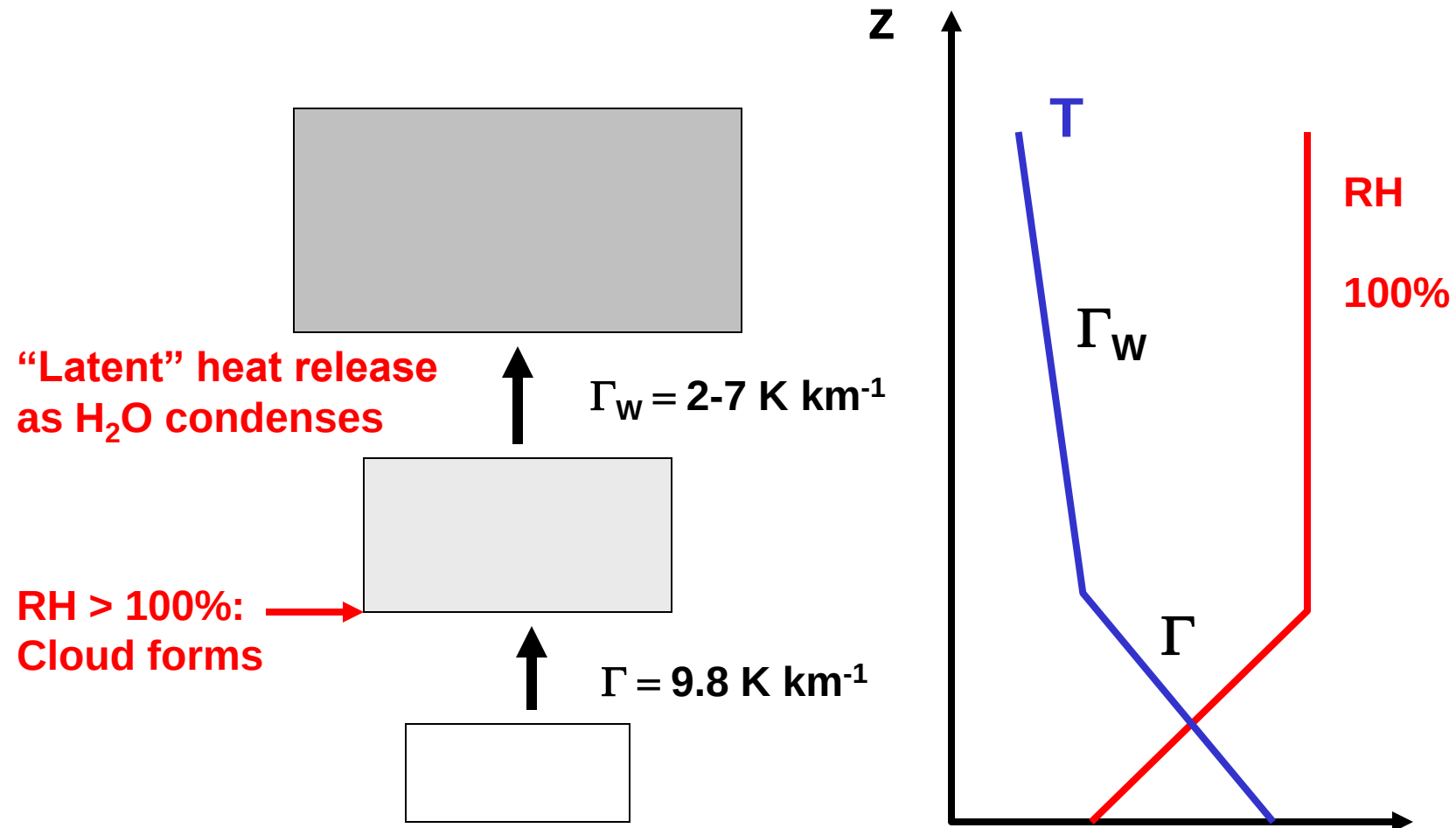
$$-\frac{dT_{ATM}}{dz} = \Gamma \quad \text{neutral}$$

$$-\frac{dT_{ATM}}{dz} < \Gamma \quad \text{stable}$$

**Particularly stable conditions are encountered when temperature increases with altitude ($dT_{ATM}/dz > 0$)
Such a condition is called temperature inversion**

IN CLOUDY AIR PARCEL, HEAT RELEASE FROM H_2O CONDENSATION MODIFIES Γ

Wet adiabatic lapse rate $\Gamma_w = 2-7 \text{ K km}^{-1}$



CLOUD FORMATION CONSEQUENCES

- **Cloud formation in a rising air parcel provides an internal source of heat that partly compensates for cooling due to expansion of the air parcel & so increases its buoyancy**
- **Although cloud formation increases buoyancy of the air parcel in which it forms, it increases the stability of the surrounding atmosphere by providing a source of heat at high altitude**
- **The dynamic of air movement through clouds causes “subsidence” which leads to a stable “P.B.L” ~1-3 km alt.**
- **Strong and persistent subsidence inversions lead to accumulation of pollutants in the PBL, over several days causing air pollution episodes**

SUBSIDENCE INVERSION

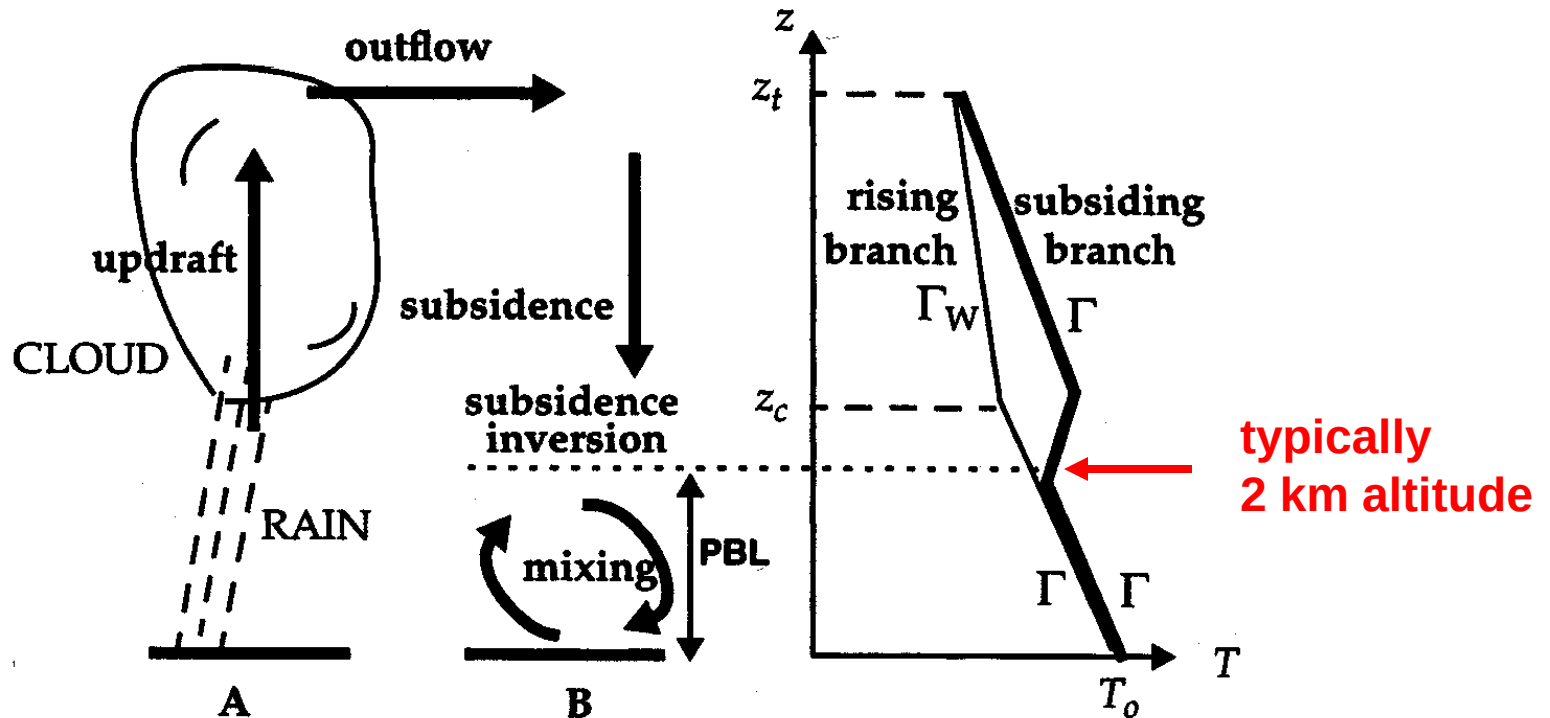
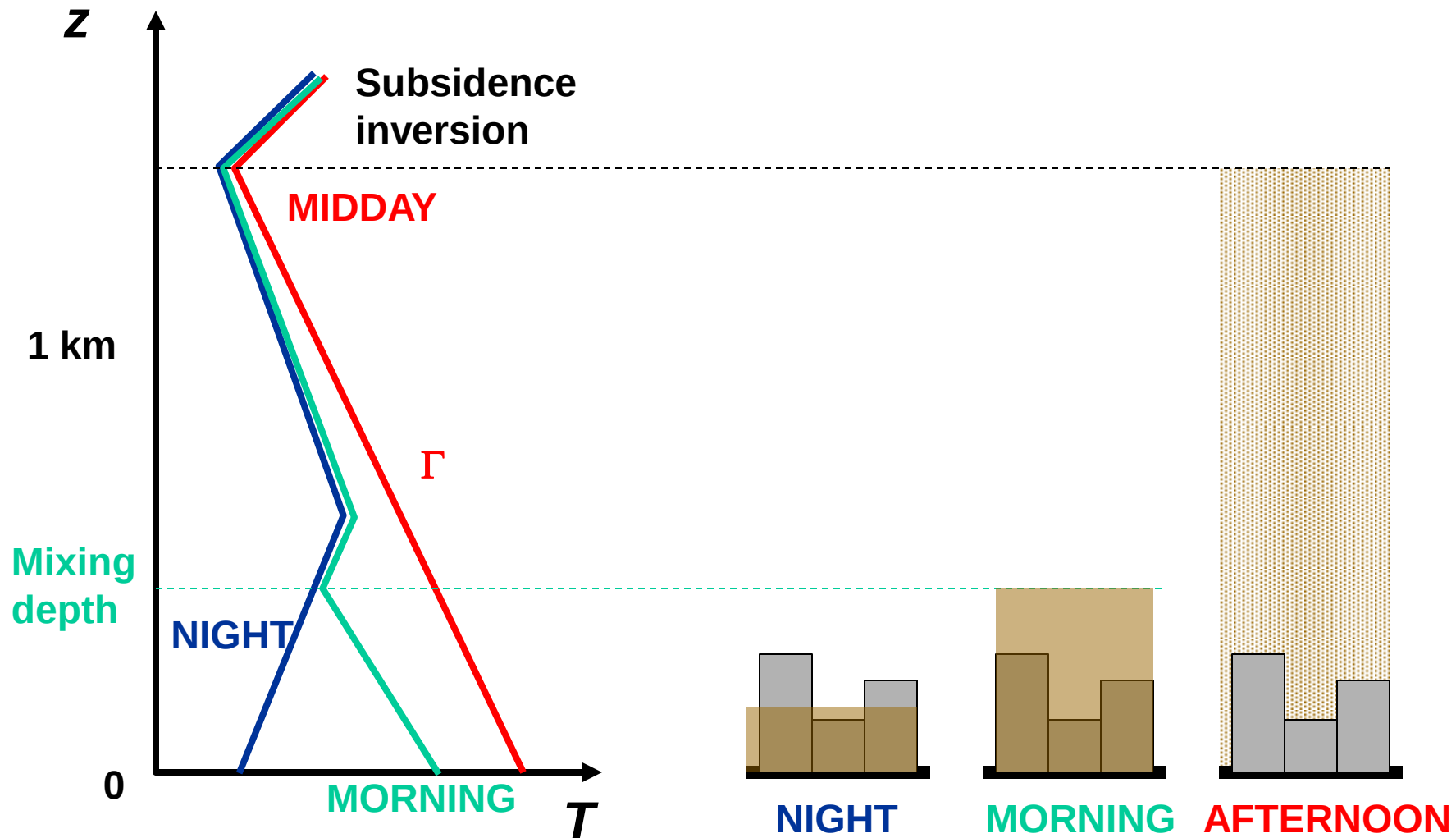
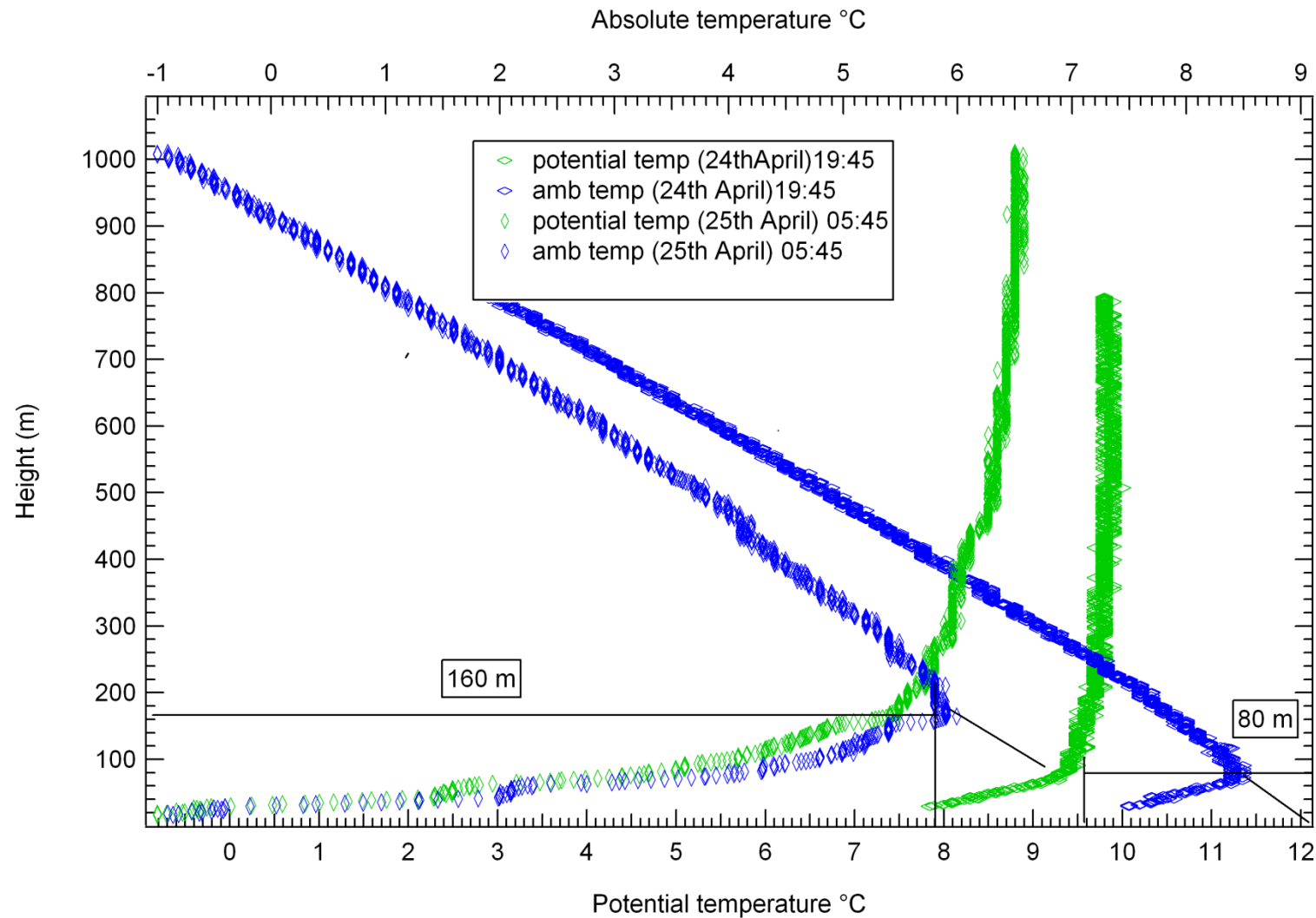


Fig. 4-17 Formation of a subsidence inversion. Temperature profiles on the right panel are shown for the upwelling region *A* (thin line) and the subsiding region *B* (bold line). It is assumed for purposes of this illustration that regions *A* and *B* have the same surface temperature T_0 . The air column extending up to the subsidence inversion is commonly called the planetary boundary layer (PBL).

DIURNAL CYCLE OF SURFACE HEATING/COOLING: ventilation of urban pollution



E.g. of Actual field data from Finland



E.g. of Stability Regimes & Boundary Layer Growth

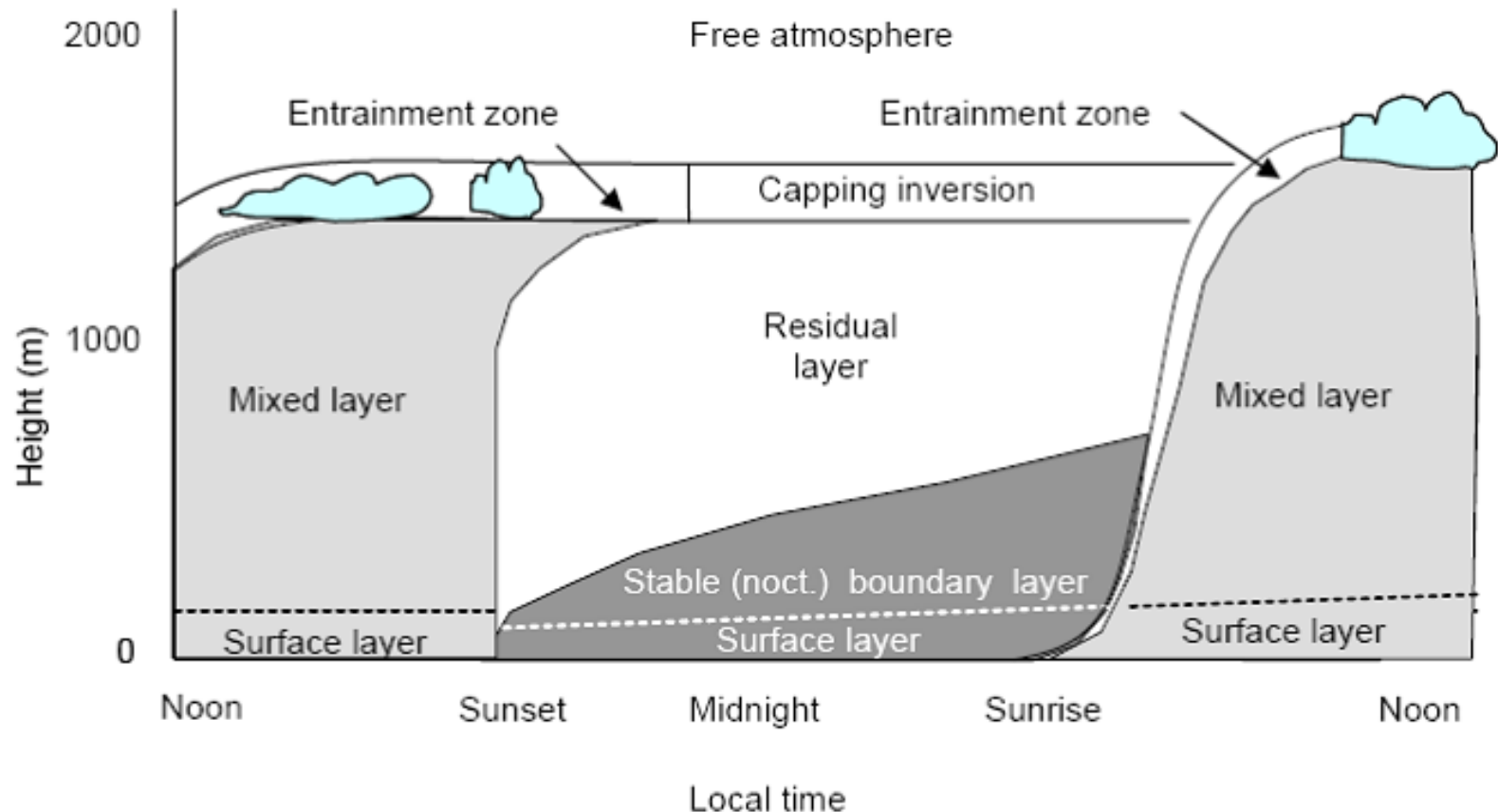


Figure 1. Diurnal development of the atmospheric boundary layer (ABL). During daytime the ABL consists of a very turbulent mixed layer (light grey). At sunset, a stable nocturnal ABL (dark grey) starts to grow from the surface upwards, and former mixed-layer air becomes a less turbulent residual layer (white). At sunrise, a new mixed layer starts to form at the surface, mixing new air into it from above at the entrainment zone. The surface layer is, by definition, the lowest 10% of the ABL. Adapted from Stull (1988)

What is Potential Temperature ?

The potential temperature of a parcel of air at pressure P is the temperature that the parcel would acquire if adiabatically brought to a standard reference pressure P_0 , usually 1000 millibars. The potential temperature is denoted by θ and, for air, is often given by:

$$\theta = T (P_0/P)^{R / C_p} \quad \text{also called "Poisson's eqn."}$$

Where T is the absolute temp (in K), R is the gas constant and C_p = specific heat capacity of air at const. pressure

$d\theta/dz = 0$ when $dT/dz = -\Gamma$, and $d\theta/dz > 0$, when atmosphere is stable; Note θ is a conserved quantity during adiabatic motions and so a very convenient indicator for atmospheric stability